Codebusters C

Montgomery HS

CSE 2025



- You will have **50 minutes** to complete this test.
- You may bring up to **(3)** 4 or 5 function calculators, if you have any additional please place it underneath your seat
- You may use the space on the exam as scratch paper, but please make sure that your answers are clear. Either write your answers separately and box them, or write them directly over/under the ciphertext.
- You may take apart the test as you wish, however you should write your team number **on top of every page** and a **250 point deduction** will be taken if the pages are **not** put back in the correct order
- The first question on the exam is **Timed**. If you believe you have solved the question in the first 10 minutes, signal to the event supervisor who will come over to check it. If you are correct, your time will be recorded and a bonus issued; if you have more than 2 errors you will be notified and allowed to retry for a bonus throughout the first 10 minutes.
- If you successfully complete the timed question after 10 minutes, you may still get the base points for the question however it will not be checked and no bonus will be awarded.
- There are 3 questions marked for **Special Bonus**: 14, 22, 24
- Page 2 contains scoring information. Pages 3 and 4 contain the reference materials. Page 5 (front of the 3rd sheet) is the **Timed Question**.

Team Name: _____

Team Number: _____

Written By:

Konnor Lee (Montgomery '26)

Scoring Table

Exam Score

Question	Points	Score
1	417	
2	213	
3	303	
4	198	
5	187	
6	268	
7	310	
8	228	
9	275	
10	203	
11	574	
12	209	

Question	Points	Score
13	214	
14	317	
15	194	
16	375	
17	248	
18	220	
19	557	
20	482	
21	309	
22	395	
23	197	
24	275	

Question	Points	Score
25	222	
26	239	
27	596	
28	323	
29	340	
30	228	
31	285	
32	262	
33	251	
Total:	9914	

Timed Question

Question Score: Circle one: 250 0

Time: _____
minutes : seconds

Timed Bonus: $(600-$_____$)\times 2=$ _____
time in seconds timed bonus

Total TQ Score: _____ $+$ _____ $=$ _____
question score timed bonus total TQ score

Special Bonus

Circle one. Special bonus is only awarded for solving the question for full score. Special bonus questions: 14, 22, 24.

- 1 Correct = 150
- 2 Correct = 400
- 3 Correct = 750
- 0 Correct = 0

Total

Final Score = _____ $+$ _____ $+$ _____ $=$ _____
Exam Score Total TQ Total Special Bonus

Rank: _____

Tables

The tables below may be useful.

Keys	a	b	c	d	e	f	g	h	i	j	k	l	m
A,B	n	o	p	q	r	s	t	u	v	w	x	y	z
C,D	o	p	q	r	s	t	u	v	w	x	y	z	n
E,F	p	q	r	s	t	u	v	w	x	y	z	n	o
G,H	q	r	s	t	u	v	w	x	y	z	n	o	p
I,J	r	s	t	u	v	w	x	y	z	n	o	p	q
K,L	s	t	u	v	w	x	y	z	n	o	p	q	r
M,N	t	u	v	w	x	y	z	n	o	p	q	r	s
O,P	u	v	w	x	y	z	n	o	p	q	r	s	t
Q,R	v	w	x	y	z	n	o	p	q	r	s	t	u
S,T	w	x	y	z	n	o	p	q	r	s	t	u	v
U,V	x	y	z	n	o	p	q	r	s	t	u	v	w
W,X	y	z	n	o	p	q	r	s	t	u	v	w	x
Y,Z	z	n	o	p	q	r	s	t	u	v	w	x	y

A	AAAAA
B	AAAAB
C	AAABA
D	AAABB
E	AABAA
F	AABAB
G	AABBA
H	AABBB
I/J	ABAAA
K	ABAAB
L	ABABA
M	ABABB

N	ABBAA
O	ABBAB
P	ABBBA
Q	ABBBB
R	BAAAA
S	BAAAB
T	BAABA
U/V	BAABB
W	BABAA
X	BABAB
Y	BABBA
Z	BABBB

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A

1	3	5	7	9	11	15	17	19	21	23	25
1	9	21	15	3	19	7	23	11	5	17	25

Frequency of English Letters

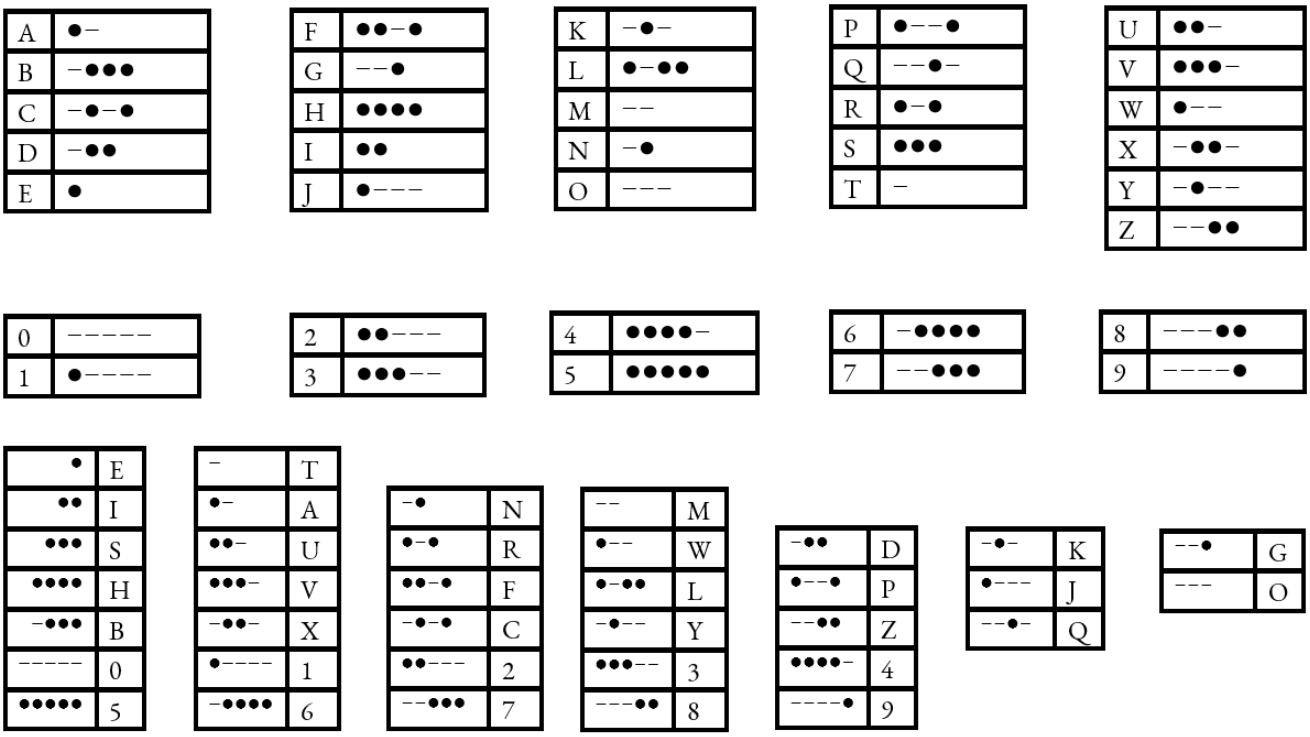
E - 12.51%	S - 6.54%	C - 3.06%	G - 1.96%	K - 0.67%
T - 9.25%	R - 6.12%	U - 2.71%	W - 1.92%	X - 0.19%
A - 8.04%	H - 5.49%	M - 2.53%	Y - 1.73%	J - 0.16%
O - 7.60%	L - 4.14%	F - 2.30%	B - 1.54%	Q - 0.11%
I - 7.26%	D - 3.99%	P - 2.00%	V - 0.99%	Z - 0.09%
N - 7.09%				

Frequency of Spanish Letters

E - 14.08%	I - 5.98%	M - 3.08%	Y - 1.09%	Z - 0.47%
A - 12.16%	L - 5.24%	P - 2.89%	V - 1.05%	Ñ - 0.17%
O - 9.20%	U - 4.69%	B - 1.49%	G - 1.00%	X - 0.14%
S - 7.20%	D - 4.67%	H - 1.18%	F - 0.69%	K - 0.11%
N - 6.83%	T - 4.60%	Q - 1.11%	J - 0.52%	W - 0.04%
R - 6.41%	C - 3.87%			

For the purposes of cryptograms it is customary to treat n and ñ as distinct letters, but a and á are the same letter. Likewise for e and é, and i and í. In other words, all the accent marks get amputated when working with cryptograms. Also, while some older Spanish dictionaries consider ch, ll, and rr, to be their own letters—this has fallen out of modern usage. Accordingly, "burro" is considered as five letters: "b-u-r-r-o" and not as four letters "b-u-rr-o."

Morse Code



Timed Question. [250 Points] Solve this **Aristocrat**. When you have finished, please meow as loudly as possible so your answer can be checked and time recorded.

YFVUL ZTBWJ VH OVIF YFVUL X RXJ TOR NFOVDS YXOOTTU;

JTD'BF UTW VU WNF HIJ XUJSTBF, YDW JTD XBFU'W CDVWF

TU WNF LBTDUR FVWNFB.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency	0	6	1	6	0	12	0	2	2	6	0	3	0	4	5	0	0	3	2	10	9	8	8	5	4	1
Replacement																										

1. (417 points) Solve this **K3 Aristocrat**. You are told that the keyword used is 9 letters long. What is the keyword? box your final answer.

NMUJSVRJ CSUJ FSPQSEBAE NAB HNPSVE, UJBZ JNI WBAZ

IOHHBABDU UJSVRJUE NCSVU UJB ESVDI SH PSSDMORJU.

K3	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency	4	7	2	3	5	1	0	4	3	9	0	0	2	5	2	3	1	3	11	0	9	5	1	0	0	2
Replacement																										

---

2. (213 points) Decode this phrase from my chemistry teacher that was encoded using the **Nihilist Substitution** cipher with a keyword of **KINETICS** and a polybius key of **EQUILIBRIUM**.

89 56 54 33 89 39 49 85 80 36 83 52 89 38 47 55

76 56 86 35 68 38 47 88 69 38 56 26 89 37 49 85

76 25 53 24 59 55 57 86 57 45 55 55 88 29 49 98

	1	2	3	4	5
1					
2					
3					
4					
5					

---

3. (303 points) Solve this **Aristocrat** from the novel Gadsby. You are told that C decodes to E.

MXN E MDQER OXYRFW OXEO E KJEYR DG PZWO YR E

ODOEAAN YRIDRWBYIZDZW WBDO YW WZIX E IEOEWOJDBXYI

EHRDJQEAYON YW XEJP OD WEN.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency	3	3	0	9	14	1	1	1	5	4	1	0	2	4	11	2	2	7	0	0	0	0	10	6	9	4
Replacement																										

4. (198 points) Decode this joke from the Daily Waffle that was encoded using the **Porta** cipher with a keyword of **STARBOARD**.

RZBFGM UQMTUR NGHEJW PQGHNX QTSAQR FEHJQM PRWMFM

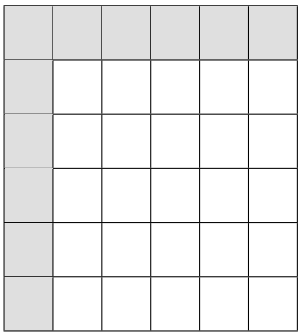
QGSRKU ZEHNJ

5. (187 points) Decode this bad science joke that was encoded using the **Checkerboard** cipher with a polybius key of **CONJUGATE**.

```
EE AW EI SL SE SS    SW CW SI SW CS CE    LS CI CW CI SW EE

AW CI SW EI SE SS    SL CE SW SL SE SS    SW CW SI SW CE SL

SI AL CW CI SW
```



---

6. (268 points) Solve this **cryptarithm** for $7498512\ 04871$. Write out your final answer and box it.

```
Base 10 Addition
Answer: 7489512 04871 (two words)

      LOG
+    RAGE
+   CURSE
---------
   SPELLS
```

---

7. (310 points) Decode this bad joke that was encoded using the **Fractionated Morse** cipher. You are told the plaintext ends with **COOL** and **0 Xs of padding** at the very end.

D A H A O R L H J Y N A H

V H K V V A B Q W O A S A L Q Q C

H K X V H K W N D

Replacement																										
	●	●	●	●	●	●	●	●	●	–	–	–	–	–	–	–	–	–	×	×	×	×	×	×	×	×
	●	●	●	–	–	–	×	×	×	●	●	●	–	–	–	×	×	×	●	●	●	–	–	–	×	×
	●	–	×	●	–	×	●	–	×	●	–	×	●	–	×	●	–	×	●	–	×	●	–	×	●	–

8. (228 points) Decode this neutral quote that was encoded using the **Hill** cipher and the encoding key **LIED**.

$$\begin{pmatrix} L & I \\ E & D \end{pmatrix} = \begin{pmatrix} 11 & 8 \\ 4 & 3 \end{pmatrix}$$

MLYPWSOVANTMFQSNWRGLQSIR

9. (275 points) Solve this **Aristocrat** about an obsolete law.

UCP NCIG YUI FHGGCVMH QCFZMCFHGI ZMCAI YUI

QCTGYHYVYHCT HG LEIJHOEI EHSI W TCCNEI WTN QWT DC

LHGY LHDUY LVYVMI QUWEEITDIG?

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency	1	0	11	3	6	3	8	10	12	1	0	4	4	3	1	1	4	0	1	6	5	4	4	0	8	2
Replacement																										

10. (203 points) Decode this important life lesson that was encoded using the **Baconian** cipher.

ATTTTTTATTATTTATAAATTTATTTTTATATTATTATTTATTTATAATTTAATA

ATTATATATTTTAAATTTTTATTATATAATTAATAATTAATATTATAATTTAATA

ATATTTATTTATTATTATTTATTTAATATTTTAAATTTTTATTATATAATTAATA

ATTAATTTAATAATA

11. (574 points) Solve this **Xenocrypt** from the novel *El guardián entre el centeno*.

EMCM BM XNBPD BDZD HÑN RDENS CN TÑP D UML UDKDGML

EMB NU I, ODSD CDXDS NU XPNCOM, CN OÑLN D ZDSUN

EMBKNSLDEPMB CPNBXSDL LN DTNPXDGD.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	Ñ	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency	0	7	6	18	5	0	2	1	1	0	2	6	9	14	3	3	6	0	1	6	2	5	0	0	5	0	2
Replacement																											

12. (209 points) Decode this phrase that was encoded using the **Porta** cipher with a keyword of **SERVICE**.

KWZSU SPFUV LVUTK PWVVU XMXFQ TVTVX KJAKN NDKEJ

EXJUG HVWVE EGWV

13. (214 points) Decode this phrase that was encoded using the **Complete Columnar** cipher. You are told that the plaintext contains **IONALLY**.

```
ITGRI LENDI FXMOO XESXA ANOXP YEXUX CLTGR XAIOO

DAELX OMX
```

---

14. (317 points) Solve this **cryptarithm** for $87430\ 532980$. Write out your final answer and box it. ★***This question is a special bonus question.***

```
Base 10 Multiplication
Answer: 87430 532980 (two words)

    WATER
*     MSU
---------
    EAEER
+  SDTUR
+ UTTRR
---------
  DAMAGER
```

---

15. (194 points) Decode this phrase that was encoded using the **Checkerboard** cipher with a polybius key of **KART**.

AW MO RR RR RL AR AL RO MR IL ML IO RR MO AW

AR MD AR AW MO ID IL AR RO RL RO RD AL RO MR

IL ML IO RR MO AW AR

---

16. (375 points) Solve this **K2 Aristocrat**. You are told that the keyword used is 7 letters long. What is the keyword? box your final answer.

EIREOI OAUCLOAAO JZN JAAOZAUBOZOTT JIO IOCLUION UZ

RINOI AR CLUMWXG IOSOJITO J TAIUZQ CLJIAOA.

Replacement																									
K2	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency	10	1	4	0	2	0	1	0	11	6	0	4	1	3	15	0	1	3	1	4	6	0	1	1	0	5

---

17. (248 points) Decode this remark about a TV that was encoded using the **Baconian** cipher.

BDUAVEDSTBAUSVTBEDUASVETBUDAVEDSTAVEDBAVEDAUSVTBEUDSTAV

BEDUAVESTBDAUVESDAVTEBUDAVEDAVESTBDAVEUDAVSTBEDAVEUDASV

TEBDAVEDAVUEDSAVETDABVEDAVUESTDABVEDUASVETBDUASTVEDABVU

ESDAVETDBAUVESDAVTEDBAVED

---

18. (220 points) Decode this phrase that was encoded using the **Nihilist Substitution** cipher with a keyword of **TURNING** and a polybius key of **DIRECTION**.

42 97 35 75 25 45 55 53 89 24 45 35 44 73 45

79 27 47 25 35 64 54 66 38 68 33 44 65 34 59

27 58 26 54 53 65 56 35

	1	2	3	4	5
1					
2					
3					
4					
5					

---

19. (557 points) Solve this **K1 Patristocrat** about a renowned university. You are told that E decodes to B.

MLFEF RRHJT KXALF TFZFH EHUAT AJTXG IUUME GRRKG

SFXUS GCEFM LGMXA LCJMJ XNGRR FOMLF WYJFM KHFFT

K1	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency	4	0	2	0	4	12	6	4	1	5	3	5	7	1	1	0	0	6	2	5	4	0	1	5	1	1
Replacement																										

---

20. (482 points) Solve this **K1 Xenocrypt** from the novel *Matar a un ruiseñor*.

CNSKLO CNS ELSDSD PA CNS SD PO ESKRORSKO MKOENKO, SÑ

ESI RS VOQSKJS PO LRSO RS CNS PO MKOENKO PO SÑQOKÑO

NÑ VAYMKS QAÑ NÑ OKYO SÑ PO YOÑA.

K1	A	B	C	D	E	F	G	H	I	J	K	L	M	N	Ñ	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency	4	0	4	3	5	0	0	0	1	1	11	3	3	8	8	19	6	3	5	19	0	0	2	0	0	3	0
Replacement																											

---

21. (309 points) Solve this **K1 Aristocrat**. You are told that the keyword used is 8 letters long. What is the keyword? box your final answer.

NOPGXO TJVD GP VMYZ DOMX, S UKOB KGZRSKQ MNGJZ

FROHSYZXD GZROX ZRMK ZRMZ FMXNGK, RDLXGQOK MKL

GCDQOK OCSYZOL.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency	0	1	2	5	0	2	8	1	0	2	8	3	7	3	10	2	3	6	4	1	1	2	0	6	3	9
Replacement																										

22. (395 points) Decode this phrase about an event that was encoded using the **Nihilist Substitution** cipher. You are told that the keyword used to encode it is between 3 and 7 letters long and characters 10-19 (starting with 47) decode to HAVEATHREE. ★***This question is a special bonus question.***

75 36 57 59 57 36 35 67 55 47 26 66 55 28 27

48 54 27 25 40 63 46 33 30 94 35 37 36 84 49

26 26 55 55 25 38 57

	1	2	3	4	5
1					
2					
3					
4					
5					

23. (197 points) Decode this phrase about a popular card game that was encoded using the **Hill** cipher and the encoding key **HAND**.

$$\begin{pmatrix} H & A \\ N & D \end{pmatrix} = \begin{pmatrix} 7 & 0 \\ 13 & 3 \end{pmatrix}$$

EAGDESHUENVXCNUCCM

---

24. (275 points) Decode this phrase about a post-Nationals activity that was encoded using the **Checkerboard** cipher. You are told that characters 27-34 (HT EG TG ET EI OH HT ET) decode to **EPISODES**. ★***This question is a special bonus question.***

EI EN HT OH EI HT ET EN EI HN ET TG TT EG ON LH

LI HG HN OG TI HN TI HI HT HT HT EG TG ET EI OH

HT ET EI OT HG HI OG HG EN HT

---

25. (222 points) Decode this partially pangramic quote that was encoded using the **Complete Columnar** cipher. You are told that the plaintext contains **TENTIO**.

UWMRN ASAXT KODAO ETHCF EYIBI SXQOU EEPUH TINPM

TNETX HBXOT NCSOE RJVTS ATR

---

26. (239 points) Solve this **Aristocrat** about performances.

KQWVW CVW KGR JRXMWP VSXWD BRV CP RVLQWDKVC: DKCVK

KRJWKQWV CPM BHPHDQ KRJWKQWV. PRIRMA LCVWD CIRSK

GQCK QCUUWPD HP IWKGWWP.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency	1	2	9	6	0	0	3	3	3	3	12	2	3	0	0	8	7	9	2	0	2	10	15	2	0	0
Replacement																										

---

27. (596 points) Solve this **K1 Patristocrat** about a sequence of events. You are told that W decodes to Y.

WINFC UVTBT SXFCA VWINX CVMBV WMBSI OXCSM CHWIN

SATZV THXMB VOISG LVTMW INYVR STMVA NFCMB TJJVH

LCHMB TMISX VS

K1	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency	3	6	7	0	0	3	1	4	7	2	0	2	9	5	2	0	0	1	8	8	1	11	5	5	1	1
Replacement																										

---

28. (323 points) Decode this phrase that was encoded using the **Fractionated Morse** cipher. You are told the plaintext contains **NEVER** corresponding to **VUBLTC where V begins with an X**.

L B W L N C B X K T T M B U K T A E

Z N I W S N V U B L T C

D U B U K Q R A N V J I Q P

R G F L J R F P

Replacement																										
	●	●	●	●	●	●	●	●	●	–	–	–	–	–	–	–	–	–	×	×	×	×	×	×	×	×
	●	●	●	–	–	–	×	×	×	●	●	●	–	–	–	×	×	×	●	●	●	–	–	–	×	×
	●	–	×	●	–	×	●	–	×	●	–	×	●	–	×	●	–	×	●	–	×	●	–	×	●	–

---

29. (340 points) Decode this recommendation that was encoded using the **Baconian** cipher. You are told that the plaintext starts with **IFYOU**.

SALVE LIMBO DECRY SHAWM FLICK FLICK ABEND ELOPE SYNOD PIPES

SPAKE NASAL TOTTY FILMS SWIMS SICKO RELAY LAMER SNORE LIMEN

ALTOS BITES CLEAN YAWPS SULFA SEMEN PILAF LUCID TROTS LACER

UNION SMOKE KNEAD RELAY SKILL GETUP WENCH SHANK FLUBS SCULL

---

30. (228 points) Solve this factual **Aristocrat**.

YH Y VPN P NGSSPK HGK CACKJ XYZC Y UCPX ZJ UKGXVCK

YE PKWPNC UPMRCXUPSS, Y WGISN UIJ XVC CEXYKC

PKWPNC.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency	1	0	11	0	2	0	4	2	2	3	8	0	1	5	0	10	0	1	5	0	5	3	3	6	7	2
Replacement																										

---

31. (285 points) Decode this play on words that was encoded using the **Porta** cipher. You are told ciphertext PIEF (Characters 34-37) decode to **INTO**.

KYXIS DKCNY KCNBX GBPEU AOWUL CBKWH WPQPI EFRUU

MXKFT LBTEB X

---

32. (262 points) Decode this phrase that was encoded using the **Complete Columnar** cipher. You are told that the plaintext contains **MULTIPLES**.

RFPLE VREUS HLUTL FAOON TEALN PTREA IVYHE TOLMO

EEOAL TPEET LYVEE SOOTO LSUFO EAEMH TI

---

33. (251 points) Decode this phrase that was encoded using the **Hill** cipher and the encoding key **EXPLOSION**.

$$\begin{pmatrix} E & X & P \\ L & O & S \\ I & O & N \end{pmatrix} = \begin{pmatrix} 4 & 23 & 15 \\ 11 & 14 & 18 \\ 8 & 14 & 13 \end{pmatrix} \qquad \begin{pmatrix} 4 & 23 & 15 \\ 11 & 14 & 18 \\ 8 & 14 & 13 \end{pmatrix}^{-1} = \begin{pmatrix} 24 & 19 & 14 \\ 3 & 4 & 19 \\ 22 & 20 & 7 \end{pmatrix}$$

AEGHZNKCMWYZBGGPSA